THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 1 & 2: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. To model a cricket shot that may go for a six, which detail is MOST important? 1

(a) the brand of the bat
(b) the mass of the ball and its speed and direction
(c) the colour of the ball
(d) the amount of grass on the field

2. The speed of light is denoted by ‘c’ because: 1

(a) c stands for centimetre
(b) c stands for current
(c) it was chosen at random by a school
(d) it comes from the Latin word celeritas meaning speed

3. In science, a ‘law’ usually: 1

(a) can never be tested
(b) explains why the pattern occurs in every case
(c) describes a regular pattern observed in nature
(d) is a guess without evidence

4. The principle of conservation of energy is an example of: 1

(a) a broad idea that helps make sense of many situations
(b) a local rule of a laboratory
(c) a unit of energy
(d) a measuring instrument

5. Weather forecasts are less reliable far into the future because: 1

(a) tiny differences in conditions grow over time
(b) instruments stop working after two days
(c) the Sun's heat stops changing
(d) air has no temperature

6. 6 g of salt is dissolved in 34 g of water. The mass by mass percentage of the solution is: 1

(a) 17.65 % **(b)** 15 % **(c)** 7.5 % **(d)** 25 %

7. 2 g of a solute is dissolved in water to make 50 mL of solution. Its mass by volume percentage (% m/v) is: 1

(a) 2 % **(b)** 4 % **(c)** 8 % **(d)** 3.85 %

8. The solubility of most solids in water generally: 1

(a) becomes zero on heating
(b) does not depend on temperature
(c) increases with temperature
(d) decreases with temperature

9. Slow cooling of a hot saturated solution gives: 1

(a) larger, well-formed crystals **(b)** a gas
(c) very small crystals **(d)** no crystals

10. Mustard oil and water can be separated using: 1

(a) a magnet **(b)** a separating funnel
(c) a chromatography strip **(d)** a condenser

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What should scientists do when a prediction does not match the observation? 2

12. Give two questions that would improve the prediction ‘it will rain this afternoon because clouds look dark’. 2

13. Name the method used to separate the fractions of crude petroleum and state why it is used. 2

14. Why is the pencil line, not an ink line, drawn at the bottom of a chromatography strip? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Check the estimate of 10 000 litres of air per day using the balloon method: one balloon of 2 litres in 20 s. 3

16. Describe how a separating funnel separates two immiscible liquids. Name the property used. 3

17. How much rice would feed a family of four for a month? Outline how to make a rough estimate. 3

OR

Why are the divisions between physics, chemistry, biology and earth science not rigid? 3

18. Explain sublimation and deposition using the example of camphor and sand. 3

OR

Differentiate between a solution, a suspension and a colloid in terms of particle size, settling and Tyndall effect. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Why do scientists make predictions? 5
(ii) Describe what they do if predictions are correct and if they fail.
(iii) Give two everyday examples of scientific predictions.

OR

(i) Explain the role of symbols, units and mathematics in the language of science. 5
(ii) Give examples of three symbols with their quantities.
(iii) Why did the speed of light get the symbol c?

20. **Case study:** A student has a mixture of sand, iron filings, camphor and common salt. 5
(i) Plan a flow chart of separation.
(ii) Which property is used for each step?
(iii) Which method recovers salt?
(iv) Which method recovers camphor?

OR

(i) What is sublimation? Give two substances that sublime. 5
(ii) Describe the activity to separate camphor and sand.
(iii) What is the reverse process called?

END OF QUESTION PAPER — **Total: 40 Marks**